

```
In [1]: #Import Libraries
import pandas as pd
import numpy as np
import xgboost as xgb
from sklearn.model_selection import train_test_split
from sklearn.model_selection import RandomizedSearchCV
from sklearn.preprocessing import RobustScaler
from sklearn.model_selection import GridSearchCV
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
In [3]: df = pd.read_csv('C:\\Users\\HP\\Desktop\\COFFEE_YIELD_PREDICTION\\coffee_yi
```

```
In [4]: #Removing extreme values
Q1 = df["Cherry Yield (Kgs)"].quantile(0.25)
Q3 = df["Cherry Yield (Kgs)"].quantile(0.75)
IQR = Q3 - Q1

# Define outlier boundaries
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

# Define target variable (Cherry Yield)
target_col = "Cherry Yield (Kgs)"

# Define features (excluding target variable)
feature_cols = [col for col in df.columns if col != target_col]

# Separate features and target
X = df[feature_cols]
y = df[target_col]

# Remove outliers
df = df[(df["Cherry Yield (Kgs)"] >= lower_bound) & (df["Cherry Yield (Kgs)"] <= upper_bound)]
# Remove rows with extreme values
df_filtered = df[(df[target_col] >= lower_bound) & (df[target_col] <= upper_bound)]

print(f"Original dataset size: {df.shape[0]} rows")
print(f"Dataset size after removing outliers: {df_filtered.shape[0]} rows")
```

Original dataset size: 384 rows

Dataset size after removing outliers: 384 rows

```
In [5]: # Cap values at lower and upper bounds
df[target_col] = np.clip(df[target_col], lower_bound, upper_bound)

print(f"Outliers have been capped between {lower_bound} and {upper_bound}.")
```

Outliers have been capped between -18831.9153562114 and 71550.3990841931.

```
In [6]: #Apply log transformation to cherry yield
df["Cherry Yield (Kgs)"] = np.log1p(df["Cherry Yield (Kgs)"])
```

```
In [7]: #Encode categorical features properly
from sklearn.preprocessing import OneHotEncoder

# Select categorical columns
categorical_cols = ["County", "Grower Type", "Coffee Variety"]

# Apply one-hot encoding
encoder = OneHotEncoder(drop="first", sparse_output=False)
encoded_features = encoder.fit_transform(df[categorical_cols])

# Convert to DataFrame
encoded_df = pd.DataFrame(encoded_features, columns=encoder.get_feature_name

# Drop original categorical columns and add encoded ones
df = df.drop(columns=categorical_cols)
df = pd.concat([df, encoded_df], axis=1)

print("Categorical features encoded successfully!")
```

Categorical features encoded successfully!

```
In [8]: print(df.columns) # Check available column names

Index(['Year', 'Rainfall (mm)', 'Temperature (°C)', 'Soil pH', 'Humidity (%)',
      'CBD Cases', 'CLR Cases', 'BBC Cases', 'Production Cost (KES)',
      'Cherry Payout (KES/kg)', 'Auction Price (USD/50Kg)',
      'Fertilizer Use (Kg/acre)', 'Insecticide Use (Liters/acre)',
      'Fungicide Use (Liters/acre)', 'Labor Cost (KES/acre)',
      'Canopy Management Score', 'Cherry Yield (Kgs)',
      'Parchment Yield (Kgs)', 'Clean Coffee Yield (Kgs)', 'County_Kirinyaga',
      'County_Murang'a', 'County_Nyeri', 'Grower Type_Estate',
      'Coffee Variety_Ruiru 11', 'Coffee Variety_SL28',
      'Coffee Variety_SL34'],
      dtype='object')
```

```
In [9]: from skopt import BayesSearchCV
from skopt.space import Real, Integer

# Define target variable
target_col = "Cherry Yield (Kgs)"

# Define feature columns (exclude target)
feature_cols = [col for col in df.columns if col != target_col]

# Split data
X = df[feature_cols]
y = df[target_col]

# Train/test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, ran

# Define XGBoost model
xgb_model = xgb.XGBRegressor(objective="reg:squarederror", enable_categorical
```

```

# Define Bayesian search space
param_space = {
    "n_estimators": Integer(50, 500),
    "learning_rate": Real(0.01, 0.3, prior="log-uniform"),
    "max_depth": Integer(3, 10),
    "subsample": Real(0.5, 1.0),
    "colsample_bytree": Real(0.5, 1.0)
}

# Bayesian Optimization
bayes_search = BayesSearchCV(
    xgb_model,
    param_space,
    n_iter=30,
    cv=5,
    scoring="r2",
    n_jobs=-1,
    random_state=42
)

# Fit model
bayes_search.fit(X_train, y_train)

# Best model from tuning
best_xgb_model_bayes = bayes_search.best_estimator_

# Predictions
y_pred_bayes = best_xgb_model_bayes.predict(X_test)

# Evaluate Model
mae_bayes = mean_absolute_error(y_test, y_pred_bayes)
mse_bayes = mean_squared_error(y_test, y_pred_bayes)
rmse_bayes = np.sqrt(mse_bayes)
r2_bayes = r2_score(y_test, y_pred_bayes)

# Print Results
print(f"Mean Absolute Error (MAE): {mae_bayes:.2f}")
print(f"Mean Squared Error (MSE): {mse_bayes:.2f}")
print(f"Root Mean Squared Error (RMSE): {rmse_bayes:.2f}")
print(f"R2 Score: {r2_bayes:.2f}")

```

Mean Absolute Error (MAE): 0.09
Mean Squared Error (MSE): 0.01
Root Mean Squared Error (RMSE): 0.11
R² Score: 0.96

```

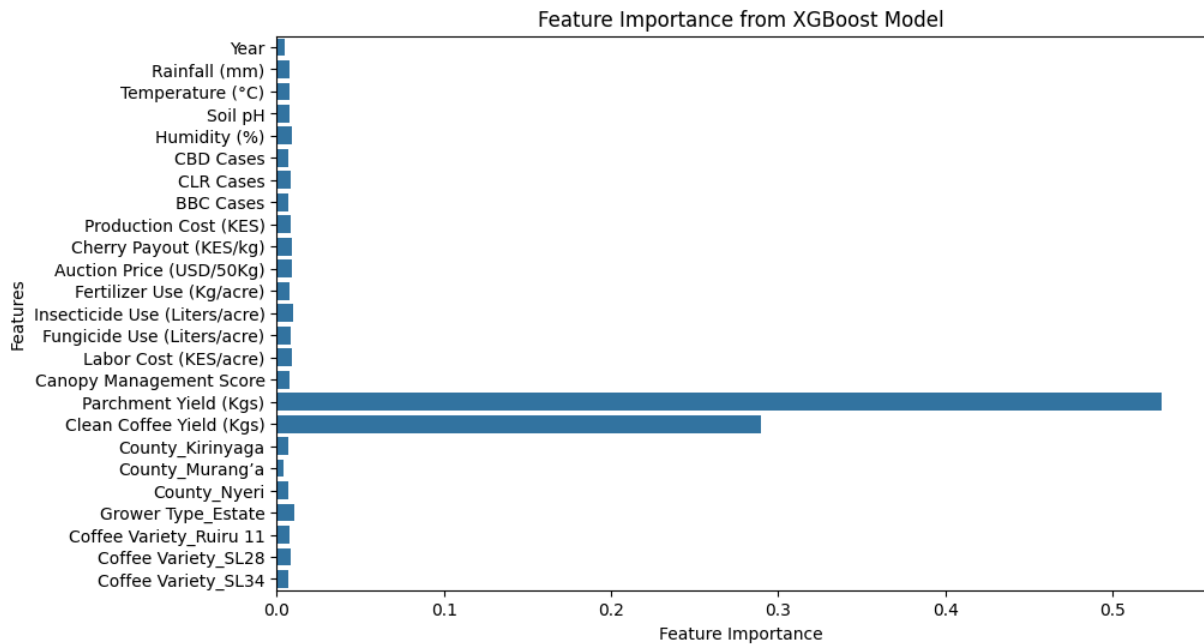
In [10]: # Get feature importance from XGBoost model
importance = best_xgb_model_bayes.feature_importances_
feature_names = X_train.columns

#import dependencies
import matplotlib.pyplot as plt
import seaborn as sns

# Plot feature importance
plt.figure(figsize=(10, 6))

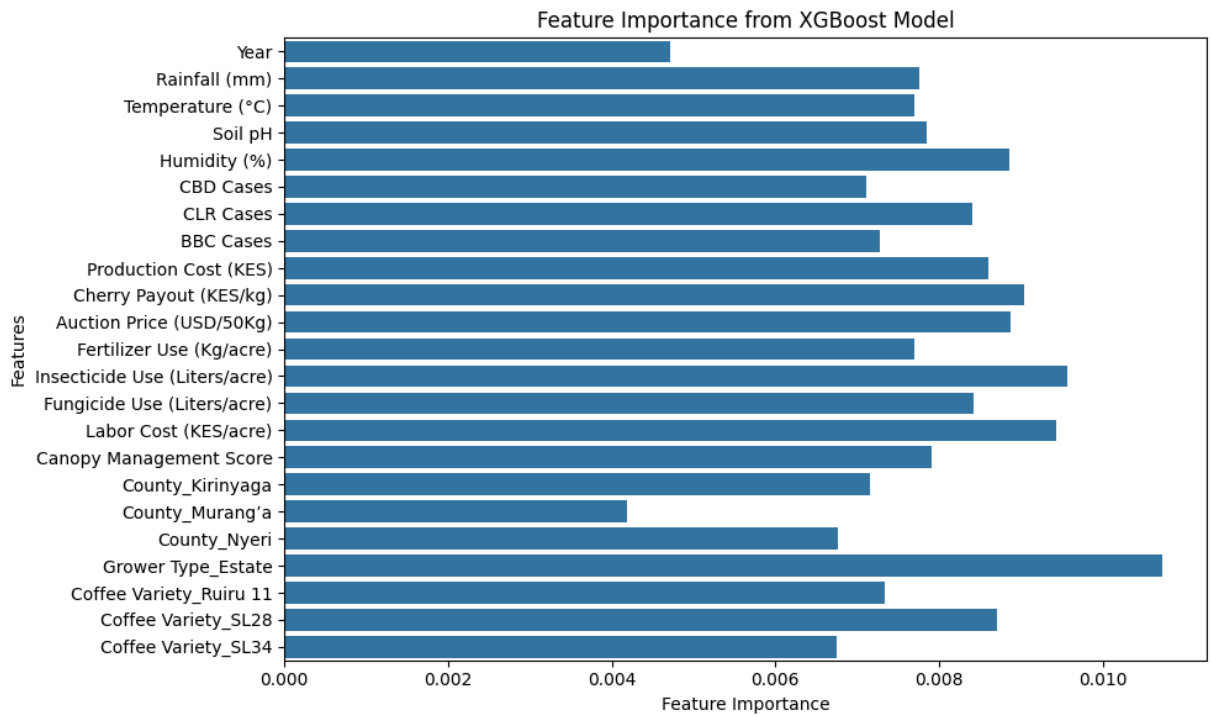
```

```
sns.barplot(x=importance, y=feature_names)
plt.xlabel("Feature Importance")
plt.ylabel("Features")
plt.title("Feature Importance from XGBoost Model")
plt.show()
```



```
In [11]: #exclude yields
excluded = ['Cherry Yield (Kgs)', 'Parchment Yield (Kgs)', 'Clean Coffee Yield (Kgs)']
filtered_features = [(f, i) for f, i in zip(feature_names, importance) if f not in excluded]

#plot feature importance
plt.figure(figsize=(10, 6))
sns.barplot(x=[i for _, i in filtered_features], y=[f for f, _ in filtered_features])
plt.xlabel("Feature Importance")
plt.ylabel("Features")
plt.title("Feature Importance from XGBoost Model")
plt.tight_layout()
plt.show()
```



```
In [12]: # Calculate the correlation matrix for the entire dataset
correlation_matrix = df.corr()

# Display full correlation matrix
print(correlation_matrix)
```

\	Year	Rainfall (mm)	Temperature (°C)
Year	1.000000e+00	-0.011327	-0.040973
Rainfall (mm)	-1.132707e-02	1.000000	-0.126625
Temperature (°C)	-4.097272e-02	-0.126625	1.000000
Soil pH	1.788839e-02	0.091005	-0.039466
Humidity (%)	1.732249e-02	-0.007394	-0.088826
CBD Cases	-1.459650e-02	-0.030073	-0.023697
CLR Cases	5.597593e-02	-0.003214	0.026307
BBC Cases	-3.678989e-02	-0.002772	0.045984
Production Cost (KES)	3.563504e-02	0.085757	-0.012898
Cherry Payout (KES/kg)	3.530770e-02	0.011071	-0.046946
Auction Price (USD/50Kg)	-8.266581e-02	0.089484	-0.039559
Fertilizer Use (Kg/acre)	-4.041553e-02	0.019957	-0.024112
Insecticide Use (Liters/acre)	4.103025e-02	0.012234	0.077799
Fungicide Use (Liters/acre)	-7.751117e-02	-0.022620	-0.020370
Labor Cost (KES/acre)	-2.580657e-02	-0.070536	0.065629
Canopy Management Score	1.962880e-02	-0.007582	0.098979
Cherry Yield (Kgs)	-4.621717e-02	-0.058241	-0.039629
Parchment Yield (Kgs)	-2.837055e-02	-0.032912	-0.025146
Clean Coffee Yield (Kgs)	-3.625326e-02	-0.034762	-0.021112
County_Kirinyaga	1.803060e-15	0.003169	0.009336
County_Murang'a	-1.663410e-15	0.032372	-0.028200
County_Nyeri	3.972456e-15	-0.086783	-0.000164
Grower Type_Estate	3.752145e-17	0.055725	0.012480
Coffee Variety_Ruiru 11	9.810253e-16	-0.012627	-0.046180
Coffee Variety_SL28	4.808417e-16	0.033483	0.060783
Coffee Variety_SL34	-3.237848e-16	0.015813	-0.010860

\	Soil pH	Humidity (%)	CBD Cases	CLR Cases
Year	0.017888	0.017322	-0.014596	0.055976
Rainfall (mm)	0.091005	-0.007394	-0.030073	-0.003214
Temperature (°C)	-0.039466	-0.088826	-0.023697	0.026307
Soil pH	1.000000	-0.029443	-0.024782	-0.012471
Humidity (%)	-0.029443	1.000000	-0.023411	0.062949
CBD Cases	-0.024782	-0.023411	1.000000	-0.085378
CLR Cases	-0.012471	0.062949	-0.085378	1.000000
BBC Cases	0.051080	-0.055816	-0.073118	-0.021247
Production Cost (KES)	-0.014593	0.024636	0.014911	-0.012900
Cherry Payout (KES/kg)	-0.056586	0.013157	-0.078909	0.139068
Auction Price (USD/50Kg)	-0.014443	-0.068419	-0.046071	0.013691
Fertilizer Use (Kg/acre)	0.000312	0.038778	-0.070169	-0.001336
Insecticide Use (Liters/acre)	-0.030618	-0.040776	0.106343	-0.030271
Fungicide Use (Liters/acre)	-0.085839	-0.027285	-0.063390	0.003159
Labor Cost (KES/acre)	-0.009218	-0.051111	0.065802	-0.025368
Canopy Management Score	0.029868	-0.002586	0.050279	-0.058606
Cherry Yield (Kgs)	-0.069340	0.016188	0.005051	-0.000674
Parchment Yield (Kgs)	-0.059154	0.054225	0.045581	0.017320
Clean Coffee Yield (Kgs)	-0.053865	0.062898	0.042028	0.020488
County_Kirinyaga	0.095088	-0.015971	-0.008177	0.002593
County_Murang'a	-0.031540	-0.104829	-0.046546	0.000265
County_Nyeri	-0.021118	0.054576	0.098754	0.069260
Grower Type_Estate	0.058433	-0.058461	-0.040855	-0.108049
Coffee Variety_Ruiru 11	0.028136	-0.027411	-0.013209	0.010635
Coffee Variety_SL28	0.083091	-0.106345	-0.040676	-0.005661

Coffee Variety_SL34	-0.044009	0.036539	0.071706	0.052540
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	BBC Cases	Production Cost (KES)	\
Year	-0.036790	0.035635	
Rainfall (mm)	-0.002772	0.085757	
Temperature (°C)	0.045984	-0.012898	
Soil pH	0.051080	-0.014593	
Humidity (%)	-0.055816	0.024636	
CBD Cases	-0.073118	0.014911	
CLR Cases	-0.021247	-0.012900	
BBC Cases	1.000000	-0.085879	
Production Cost (KES)	-0.085879	1.000000	
Cherry Payout (KES/kg)	-0.022388	0.129900	
Auction Price (USD/50Kg)	-0.034219	-0.022765	
Fertilizer Use (Kg/acre)	-0.035860	0.003568	
Insecticide Use (Liters/acre)	0.056691	-0.017553	
Fungicide Use (Liters/acre)	0.037723	0.061037	
Labor Cost (KES/acre)	0.013798	-0.123770	
Canopy Management Score	0.066798	-0.032759	
Cherry Yield (Kgs)	0.017545	0.044850	
Parchment Yield (Kgs)	0.024758	0.063147	
Clean Coffee Yield (Kgs)	0.021073	0.063166	
County_Kirinyaga	-0.003192	-0.030860	
County_Murang'a	0.003398	0.137159	
County_Nyeri	-0.001957	-0.005297	
Grower Type_Estate	0.012129	-0.053443	
Coffee Variety_Ruiru 11	-0.089080	0.049576	
Coffee Variety_SL28	0.048299	0.022020	
Coffee Variety_SL34	0.076928	-0.038615	

	Cherry Payout (KES/kg)	...	\
Year	0.035308	...	
Rainfall (mm)	0.011071	...	
Temperature (°C)	-0.046946	...	
Soil pH	-0.056586	...	
Humidity (%)	0.013157	...	
CBD Cases	-0.078909	...	
CLR Cases	0.139068	...	
BBC Cases	-0.022388	...	
Production Cost (KES)	0.129900	...	
Cherry Payout (KES/kg)	1.000000	...	
Auction Price (USD/50Kg)	0.060499	...	
Fertilizer Use (Kg/acre)	-0.036403	...	
Insecticide Use (Liters/acre)	0.062078	...	
Fungicide Use (Liters/acre)	0.017377	...	
Labor Cost (KES/acre)	0.004358	...	
Canopy Management Score	-0.019105	...	
Cherry Yield (Kgs)	0.010587	...	
Parchment Yield (Kgs)	0.010926	...	
Clean Coffee Yield (Kgs)	0.009700	...	
County_Kirinyaga	0.010818	...	
County_Murang'a	0.050791	...	
County_Nyeri	-0.107959	...	
Grower Type_Estate	-0.019291	...	
Coffee Variety_Ruiru 11	-0.005641	...	
Coffee Variety_SL28	-0.062996	...	

Coffee Variety_SL34

0.026733 ...

	Cherry Yield (Kgs)	Parchment Yield (Kgs) \
Year	-0.046217	-0.028371
Rainfall (mm)	-0.058241	-0.032912
Temperature (°C)	-0.039629	-0.025146
Soil pH	-0.069340	-0.059154
Humidity (%)	0.016188	0.054225
CBD Cases	0.005051	0.045581
CLR Cases	-0.000674	0.017320
BBC Cases	0.017545	0.024758
Production Cost (KES)	0.044850	0.063147
Cherry Payout (KES/kg)	0.010587	0.010926
Auction Price (USD/50Kg)	-0.002505	-0.017629
Fertilizer Use (Kg/acre)	-0.114084	-0.124734
Insecticide Use (Liters/acre)	0.056172	0.052851
Fungicide Use (Liters/acre)	0.062918	0.053528
Labor Cost (KES/acre)	0.001623	0.005850
Canopy Management Score	0.029999	0.065963
Cherry Yield (Kgs)	1.000000	0.930893
Parchment Yield (Kgs)	0.930893	1.000000
Clean Coffee Yield (Kgs)	0.928177	0.997445
County_Kirinyaga	0.014500	-0.003932
County_Murang'a	0.020231	0.020230
County_Nyeri	-0.020761	-0.003966
Grower Type_Estate	-0.006454	-0.037997
Coffee Variety_Ruiru 11	0.070879	0.065934
Coffee Variety_SL28	-0.092281	-0.080667
Coffee Variety_SL34	0.011353	0.025104

	Clean Coffee Yield (Kgs)	County_Kirinyaga \
Year	-0.036253	1.803060e-15
Rainfall (mm)	-0.034762	3.168667e-03
Temperature (°C)	-0.021112	9.336180e-03
Soil pH	-0.053865	9.508771e-02
Humidity (%)	0.062898	-1.597136e-02
CBD Cases	0.042028	-8.177056e-03
CLR Cases	0.020488	2.592630e-03
BBC Cases	0.021073	-3.192471e-03
Production Cost (KES)	0.063166	-3.085952e-02
Cherry Payout (KES/kg)	0.009700	1.081821e-02
Auction Price (USD/50Kg)	-0.012751	5.831892e-04
Fertilizer Use (Kg/acre)	-0.118662	4.226529e-02
Insecticide Use (Liters/acre)	0.049219	1.105502e-01
Fungicide Use (Liters/acre)	0.054314	-1.121164e-01
Labor Cost (KES/acre)	0.002418	-1.556506e-02
Canopy Management Score	0.054422	8.811683e-02
Cherry Yield (Kgs)	0.928177	1.450007e-02
Parchment Yield (Kgs)	0.997445	-3.931636e-03
Clean Coffee Yield (Kgs)	1.000000	-4.361901e-03
County_Kirinyaga	-0.004362	1.000000e+00
County_Murang'a	0.013709	-3.333333e-01
County_Nyeri	0.001309	-3.333333e-01
Grower Type_Estate	-0.037109	6.343106e-18
Coffee Variety_Ruiru 11	0.067876	-2.698459e-18
Coffee Variety_SL28	-0.078168	-1.503427e-17

Coffee Variety_SL34	0.021673	-1.079383e-17
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	County_Murang'a	County_Nyeri	\
Year	-1.663410e-15	3.972456e-15	
Rainfall (mm)	3.237239e-02	-8.678258e-02	
Temperature (°C)	-2.820048e-02	-1.643655e-04	
Soil pH	-3.154038e-02	-2.111751e-02	
Humidity (%)	-1.048294e-01	5.457608e-02	
CBD Cases	-4.654632e-02	9.875368e-02	
CLR Cases	2.645541e-04	6.926027e-02	
BBC Cases	3.398437e-03	-1.956676e-03	
Production Cost (KES)	1.371590e-01	-5.297429e-03	
Cherry Payout (KES/kg)	5.079124e-02	-1.079594e-01	
Auction Price (USD/50Kg)	3.493403e-03	9.390034e-03	
Fertilizer Use (Kg/acre)	-1.231618e-01	3.310211e-02	
Insecticide Use (Liters/acre)	-1.149041e-01	1.158506e-03	
Fungicide Use (Liters/acre)	4.226376e-02	8.937632e-03	
Labor Cost (KES/acre)	-1.021891e-02	2.538976e-02	
Canopy Management Score	-1.240634e-02	-6.176399e-02	
Cherry Yield (Kgs)	2.023086e-02	-2.076052e-02	
Parchment Yield (Kgs)	2.022962e-02	-3.966376e-03	
Clean Coffee Yield (Kgs)	1.370903e-02	1.309311e-03	
County_Kirinyaga	-3.333333e-01	-3.333333e-01	
County_Murang'a	1.000000e+00	-3.333333e-01	
County_Nyeri	-3.333333e-01	1.000000e+00	
Grower Type_Estate	-8.012345e-18	1.034928e-17	
Coffee Variety_Ruiru 11	-6.167906e-18	3.854941e-19	
Coffee Variety_SL28	1.619075e-17	8.095376e-18	
Coffee Variety_SL34	4.356083e-17	7.709882e-18	

	Grower Type_Estate	Coffee Variety_Ruiru 11	
\			
Year	3.752145e-17	9.810253e-16	
Rainfall (mm)	5.572513e-02	-1.262712e-02	
Temperature (°C)	1.247956e-02	-4.617969e-02	
Soil pH	5.843334e-02	2.813637e-02	
Humidity (%)	-5.846103e-02	-2.741139e-02	
CBD Cases	-4.085503e-02	-1.320909e-02	
CLR Cases	-1.080486e-01	1.063508e-02	
BBC Cases	1.212927e-02	-8.908024e-02	
Production Cost (KES)	-5.344262e-02	4.957621e-02	
Cherry Payout (KES/kg)	-1.929113e-02	-5.640894e-03	
Auction Price (USD/50Kg)	-4.111017e-02	-8.694761e-03	
Fertilizer Use (Kg/acre)	-1.764435e-02	4.406635e-02	
Insecticide Use (Liters/acre)	8.539785e-02	4.283395e-02	
Fungicide Use (Liters/acre)	3.798363e-02	1.143044e-02	
Labor Cost (KES/acre)	6.653064e-02	4.018310e-02	
Canopy Management Score	-4.164639e-02	1.922111e-02	
Cherry Yield (Kgs)	-6.453643e-03	7.087911e-02	
Parchment Yield (Kgs)	-3.799692e-02	6.593425e-02	
Clean Coffee Yield (Kgs)	-3.710939e-02	6.787565e-02	
County_Kirinyaga	6.343106e-18	-2.698459e-18	
County_Murang'a	-8.012345e-18	-6.167906e-18	
County_Nyeri	1.034928e-17	3.854941e-19	
Grower Type_Estate	1.000000e+00	-5.341563e-18	
Coffee Variety_Ruiru 11	-5.341563e-18	1.000000e+00	

Coffee Variety_SL28	1.335391e-17	-3.333333e-01
Coffee Variety_SL34	2.270164e-17	-3.333333e-01
	Coffee Variety_SL28	Coffee Variety_SL34
Year	4.808417e-16	-3.237848e-16
Rainfall (mm)	3.348320e-02	1.581322e-02
Temperature (°C)	6.078306e-02	-1.086041e-02
Soil pH	8.309087e-02	-4.400890e-02
Humidity (%)	-1.063452e-01	3.653865e-02
CBD Cases	-4.067561e-02	7.170650e-02
CLR Cases	-5.661458e-03	5.254045e-02
BBC Cases	4.829900e-02	7.692825e-02
Production Cost (KES)	2.201972e-02	-3.861462e-02
Cherry Payout (KES/kg)	-6.299646e-02	2.673262e-02
Auction Price (USD/50Kg)	6.344068e-03	-5.924216e-02
Fertilizer Use (Kg/acre)	2.089994e-02	3.019372e-02
Insecticide Use (Liters/acre)	-2.132847e-03	-6.870621e-03
Fungicide Use (Liters/acre)	-1.255810e-02	-4.642921e-03
Labor Cost (KES/acre)	-2.261735e-02	6.720333e-02
Canopy Management Score	1.200522e-02	-4.531529e-02
Cherry Yield (Kgs)	-9.228060e-02	1.135272e-02
Parchment Yield (Kgs)	-8.066708e-02	2.510412e-02
Clean Coffee Yield (Kgs)	-7.816774e-02	2.167256e-02
County_Kirinyaga	-1.503427e-17	-1.079383e-17
County_Murang'a	1.619075e-17	4.356083e-17
County_Nyeri	8.095376e-18	7.709882e-18
Grower Type_Estate	1.335391e-17	2.270164e-17
Coffee Variety_Ruiru 11	-3.333333e-01	-3.333333e-01
Coffee Variety_SL28	1.000000e+00	-3.333333e-01
Coffee Variety_SL34	-3.333333e-01	1.000000e+00

[26 rows x 26 columns]

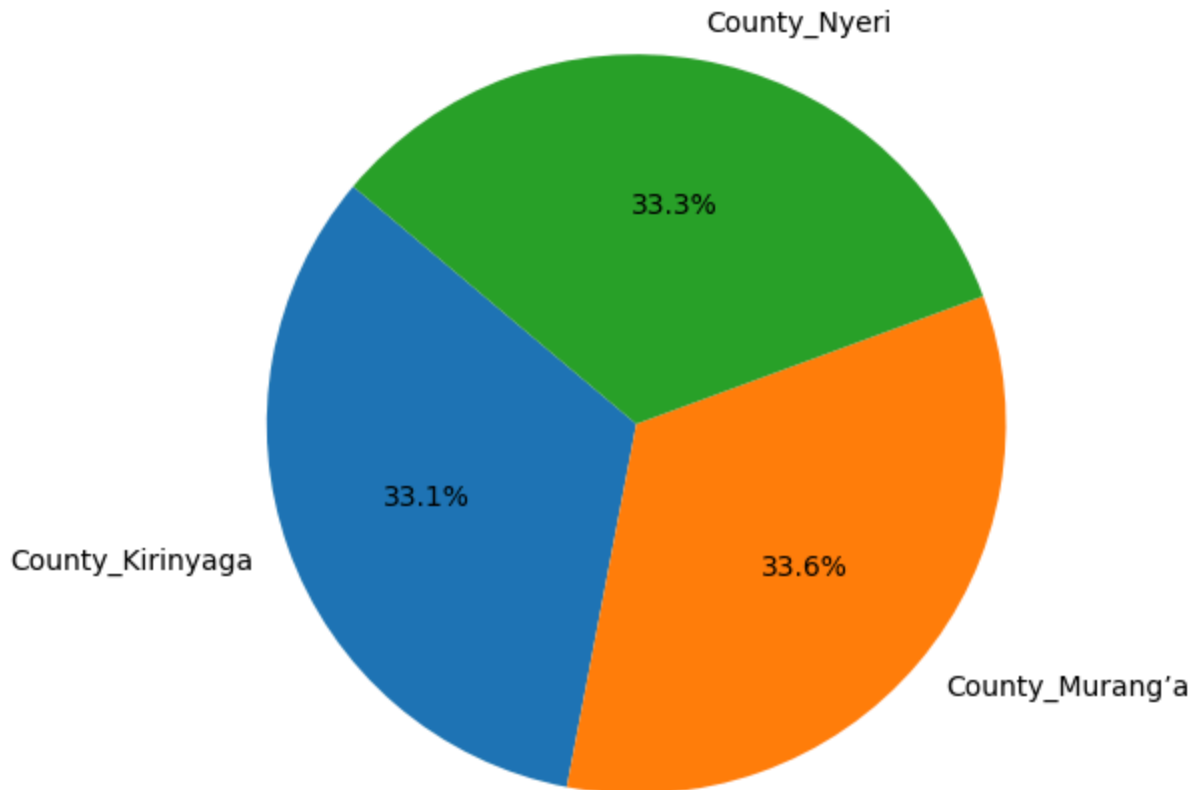
```
In [13]: import matplotlib.pyplot as plt

# Create a list of the counties am interested in
counties = ['County_Kirinyaga', 'County_Murang'a', 'County_Nyeri']

# Calculate the total Clean Coffee Yield for each county
county_clean_coffee_yield = df[counties].apply(lambda x: df['Clean Coffee Yi

# Pie chart for Clean Coffee Yield by County
plt.figure(figsize=(6, 6))
plt.pie(county_clean_coffee_yield, labels=county_clean_coffee_yield.index, a
plt.title('Distribution of Clean Coffee Yield by County')
plt.show()
```

Distribution of Clean Coffee Yield by County



```
In [14]: # Check the 'Year' column directly to see what data it contains
print(df['Year'].head())
```

```
0    2013
1    2013
2    2013
3    2013
4    2013
Name: Year, dtype: int64
```

```
In [15]: # First, make sure that 'Date' is in datetime format, if it exists
if 'Date' in df.columns:
    df['Date'] = pd.to_datetime(df['Date'], errors='coerce') # Coerce invalid
    df['Year'] = df['Date'].dt.year # Extract the year
else:
    # If 'Date' column is not there, ensure 'Year' is directly an integer and
    df['Year'] = df['Year'].astype(int)

# Check the first few rows to make sure 'Year' is fixed
print(df['Year'].head())
```

```
0    2013
1    2013
2    2013
3    2013
4    2013
Name: Year, dtype: int64
```

```
In [16]: # Example of manually fixing year if required
df['Year'] = df['Year'] // 1000000000 # Remove the extra time-related digit

# Check the 'Year' column after fixing it
print(df['Year'].head())
```

```
0    0
1    0
2    0
3    0
4    0
Name: Year, dtype: int64
```

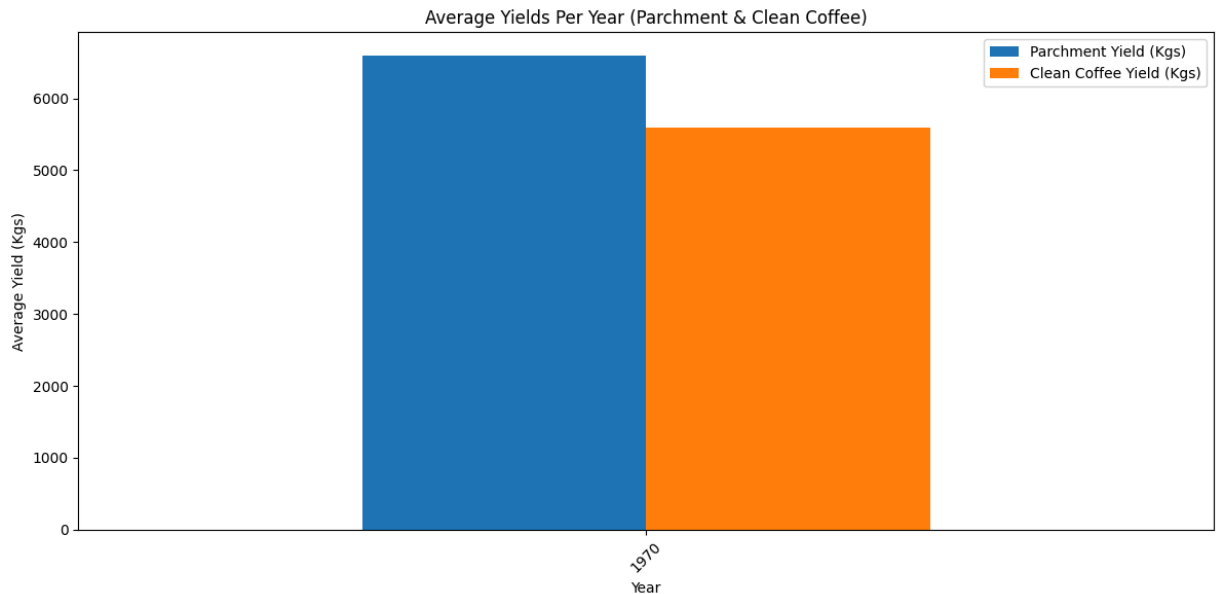
```
In [17]: # Convert the Unix timestamp (seconds since 1970) to datetime
df['Year'] = pd.to_datetime(df['Year'], unit='s').dt.year # 'unit=s' indicates seconds

# Check the first few rows to ensure it's correct
print(df['Year'].head())
```

```
0    1970
1    1970
2    1970
3    1970
4    1970
Name: Year, dtype: int32
```

```
In [18]: # Group by 'Year' and calculate the average of Parchment and Clean Coffee Yields
avg_yields_per_year = df.groupby('Year')[['Parchment Yield (Kgs)', 'Clean Coffee Yield (Kgs)']]

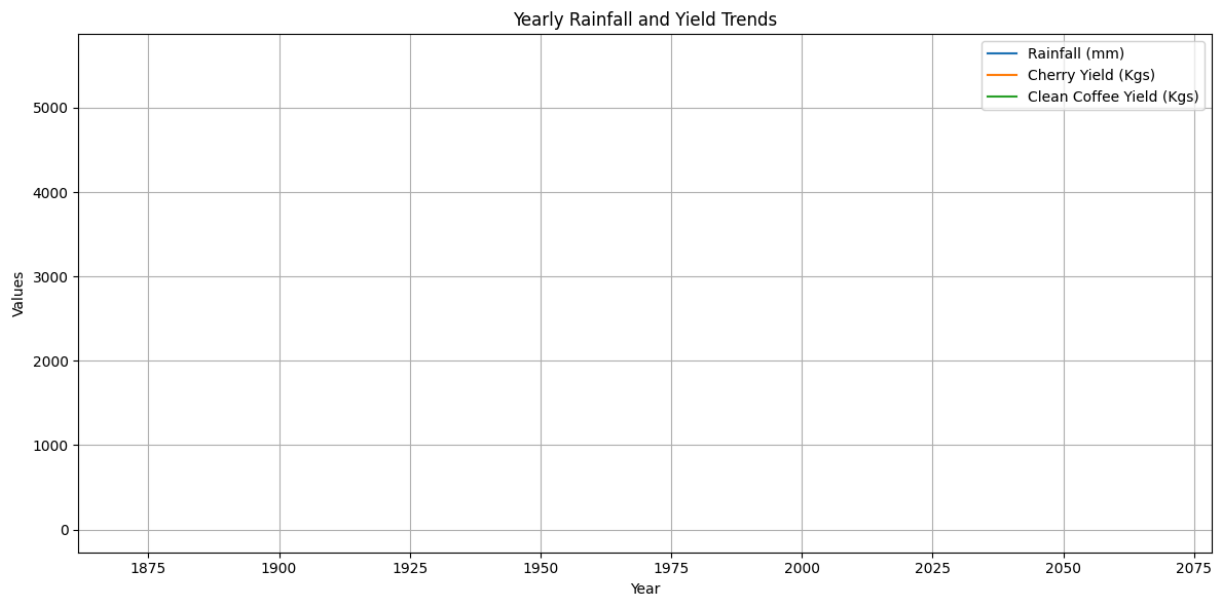
# Plot the average yields per year for Parchment and Clean Coffee only
avg_yields_per_year.plot(kind='bar', figsize=(12, 6)) # 'bar' for vertical bars
plt.title("Average Yields Per Year (Parchment & Clean Coffee)")
plt.xlabel("Year")
plt.ylabel("Average Yield (Kgs)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```
In [19]: # Ensure 'Year' is in datetime format
df['Year'] = pd.to_datetime(df['Year'], format='%Y')

# Group by year and calculate average rainfall and yield
yearly_data = df.groupby(df['Year'].dt.year)[['Rainfall (mm)', 'Cherry Yield (Kgs)', 'Clean Coffee Yield (Kgs)']]

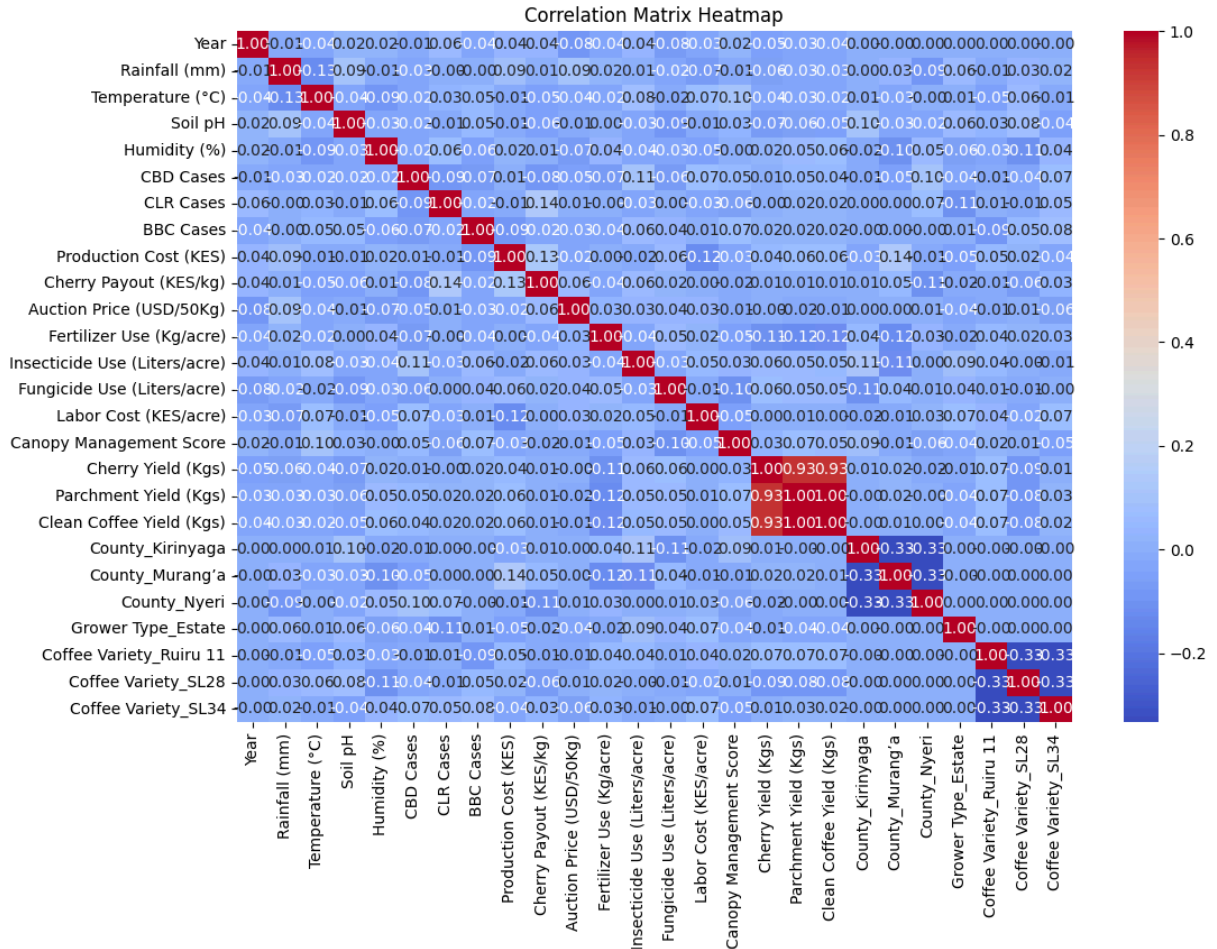
# Plot line graph
yearly_data.plot(figsize=(12, 6))
plt.title('Yearly Rainfall and Yield Trends')
plt.xlabel('Year')
plt.ylabel('Values')
plt.grid(True)
plt.tight_layout()
plt.show()
```



```
In [20]: import seaborn as sns

# Plot correlation heatmap
```

```
plt.figure(figsize=(12, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Matrix Heatmap')
plt.show()
```



```
In [21]: import joblib
```

```
# Save the model
joblib.dump(best_xgb_model_bayes, 'best_xgb_model_bayes.pkl')
print("Model saved successfully as 'best_xgb_model_bayes.pkl'")
```

Model saved successfully as 'best_xgb_model_bayes.pkl'

```
In [22]: import os
```

```
print("Current working directory:", os.getcwd())
```

Current working directory: c:\Users\HP\Desktop\COFFEE_YIELD_PREDICTION